Enterprise AI Architect Learning Path v3 – Phase Guide 2

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Azure AI Foundry & RAG Foundations

Purpose of This Phase

Phase 1 established the engineering foundation required to build reliable AI systems.

Phase 2 introduces the first true AI architecture:

> Retrieval-Augmented Generation (RAG)

This phase focuses on building a production-quality knowledge retrieval platform using Azure AI Foundry, Azure OpenAI, and Azure AI Search.

The objective is not simply to connect an LLM to documents. The objective is to build a system that produces grounded, traceable, and trustworthy responses.

By the end of this phase, the RPG Catalog evolves from a collection of files into a searchable knowledge platform capable of supporting later agent-based systems.

Why This Phase Matters

Most AI applications fail because they rely entirely on model memory.

Enterprise systems require:

- Grounded responses
- Explainable outputs
- Traceable sources
- Updatable knowledge
- Search and retrieval controls

RAG solves these problems by separating:

Knowledge

The information stored in documents.

Retrieval

The process of finding relevant information.

Reasoning

The language model's ability to synthesize and explain retrieved information.

This architectural separation becomes a recurring pattern throughout the remainder of the learning path.

Phase Objectives

Upon completion of this phase, you should be able to:

- Understand embeddings and vector search
- Build Azure AI Search indexes
- Configure Azure AI Foundry resources
- Generate document embeddings
- Design chunking strategies
- Implement hybrid search
- Design retrieval APIs
- Create grounded responses
- Implement source citations
- Build the first production-ready AI architecture

Architect Mindset

Architects should think of RAG systems as information retrieval platforms first and AI systems second.

The primary questions become:

- How is knowledge organized?
- How is relevance determined?
- How are results validated?
- How is trust established?

The model is only one component of the architecture.

Primary Resources

Required

R3

Microsoft Foundry Documentation

Focus:

- Projects

- Models
- Agents
- Evaluations
- SDKs

R4

Azure AI Foundry Learning Path

Focus:

- Foundry workflows
- Agent services
- Evaluations
- Guardrails

R5

Azure AI Search Documentation

Focus:

- Search indexes
- Queries
- Search architecture

R6

Azure AI Search Vector Search Overview

Focus:

- Embeddings
- Similarity search
- Hybrid retrieval

R7

Azure AI Search Vector Samples

Focus:

- Reference implementations
- Python examples
- Azure SDK patterns

Supporting Resources

R1

AI Engineering

Focus:

- Chunking
- Retrieval quality
- RAG design patterns

R19

AI on Azure

Focus:

- Azure OpenAI
- Azure deployment models
- AI Search integration

Core Technical Skills

1. Embeddings

Learn:

- What an embedding is
- Why embeddings work
- Similarity matching
- Semantic search

Understand:

> Embeddings convert text into vector representations that allow semantic comparison.

The architecture matters more than the mathematics.

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2. Chunking Strategies

Documents must be broken into searchable units.

Learn:

- Fixed-size chunking
- Sliding-window chunking
- Semantic chunking
- Metadata-aware chunking

Questions to answer:

- How large should chunks be?
- How much overlap should exist?
- How are citations preserved?

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3. Azure AI Search

Learn:

- Search indexes
- Index schemas
- Indexers
- Documents
- Filters
- Ranking

Understand:

Azure AI Search becomes the primary retrieval engine for future phases.

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4. Hybrid Search

Understand the difference between:

Keyword Search

Traditional term matching.

Vector Search

Semantic similarity.

Hybrid Search

Combines both approaches.

Enterprise systems frequently use hybrid search because it provides:

- Better precision
- Better recall
- Better explainability

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5. Azure AI Foundry

Learn:

- Projects

- Model deployments
- Prompt testing
- Evaluations
- Monitoring capabilities

This becomes the central AI development platform used throughout later phases.

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6. Retrieval API Design

Build APIs that:

- Accept questions
- Retrieve content
- Generate grounded responses
- Return citations

Focus on reliability and maintainability rather than frontend experiences.

Phase Project

RPG Catalog RAG v0.5

This project transforms the RPG Catalog from a file repository into a retrieval platform.

Functional Requirements

Ingest Content

Read normalized content from the Phase 1 ingestion pipeline.

Generate Embeddings

Create vector representations for:

- Rules
- Lore
- World information
- Adventure content

Build Search Index

Store:

- Chunks
- Metadata
- Embeddings

within Azure AI Search.

Retrieve Information

Support:

- Semantic search
- Keyword search
- Hybrid search

Generate Responses

Use Azure OpenAI to create:

- Grounded answers
- Search-assisted responses
- Source-aware outputs

Provide Citations

Every answer should include references to source materials.

A response without source attribution should be considered incomplete.

Architecture Deliverables

ADR-004

Chunking Strategy

Document:

- Chunk size
- Overlap strategy
- Metadata preservation

ADR-005

Retrieval Strategy

Document:

- Vector search
- Hybrid search
- Ranking approach

ADR-006

Citation Contract

Document:

- Citation format
- Source attribution expectations
- Retrieval confidence considerations

Portfolio Artifacts

C4 Context Diagram

Show:

- User
- Retrieval API
- Azure OpenAI
- Azure AI Search
- Source Content

C4 Container Diagram

Show:

- Ingestion Pipeline
- Search Index
- Retrieval Service
- Model Service

Search Architecture Document

Describe:

- Data flow
- Embedding generation
- Search process
- Response generation

Retrieval Testing Notes

Capture:

- Example queries
- Search behavior
- Observations
- Future improvements

Suggested Sprint Progression

Sprint 1 — Azure Foundation

Goals:

- Foundry setup
- Resource exploration
- Azure Search setup

Done When:

- Cloud resources operational

Sprint 2 — Embeddings

Goals:

- Embedding generation
- Document processing
- Experimentation

Done When:

- Vector generation working

Sprint 3 — Search Index

Goals:

- Schema design

- Document loading
- Metadata storage

Done When:

- Search index populated

Sprint 4 — Retrieval

Goals:

- Search APIs
- Query execution
- Result ranking

Done When:

- Search results returned successfully

Sprint 5 — Generation Layer

Goals:

- Azure OpenAI integration
- Grounded prompts
- Citation support

Done When:

- Basic RAG responses working

Sprint 6 — Hardening

Goals:

- Testing
- Logging
- Performance reviews
- Documentation updates

Done When:

- Stable RAG v0.5 platform completed

Common Pitfalls

Pitfall 1

Treating Retrieval and Generation as the Same System.

They are separate architectural concerns.

A weak retrieval layer produces weak answers regardless of model quality.

Pitfall 2

Obsessing Over Models

Many beginners repeatedly swap LLMs.

Most quality improvements actually come from:

- Better chunking
- Better metadata
- Better retrieval

Pitfall 3

Skipping Citations

Enterprise users trust systems that can demonstrate where answers originated.

Citations should exist from the beginning.

Pitfall 4

Ignoring Metadata

Metadata often becomes more valuable than embeddings.

Examples:

- System
- Rule Set
- Source Book
- Publication
- Category
- Tags

Good metadata dramatically improves retrieval quality.

Enterprise Translation Layer

The RPG Catalog project represents a reusable enterprise pattern.

Possible enterprise equivalents include:

	Portfolio Project	Enterprise Equivalent
1	RPG Catalog RAG	Legal Policy Assistant
2	RPG Catalog RAG	Technical Documentation Search
3	RPG Catalog RAG	Knowledge Management Platform
4	RPG Catalog RAG	Regulatory Compliance Assistant
5	RPG Catalog RAG	Internal Help Desk Knowledge Base

The underlying architecture remains largely identical.

Definition of Done

Phase 2 is complete when:

- Azure AI Foundry environment is operational
- Azure AI Search index is operational
- Embeddings are generated
- Content is indexed
- Retrieval APIs function correctly
- Grounded responses are generated
- Citations are returned
- Architecture diagrams exist
- ADRs are documented
- Retrieval testing has been completed

End-of-Phase Self-Assessment

You are ready for Phase 3 when you can confidently answer:

- Do I understand how embeddings and vector search work?
- Can I design and maintain an Azure AI Search index?
- Can I explain the difference between keyword, vector, and hybrid search?
- Can I generate grounded responses with source citations?
- Can I evaluate retrieval quality independently of model quality?
- Can I defend my chunking and indexing decisions?

If all answers are "yes," proceed to **Phase 3 – Production RAG & Evaluation**.